

Subject:
Biology

System:
Digestion and Excretion

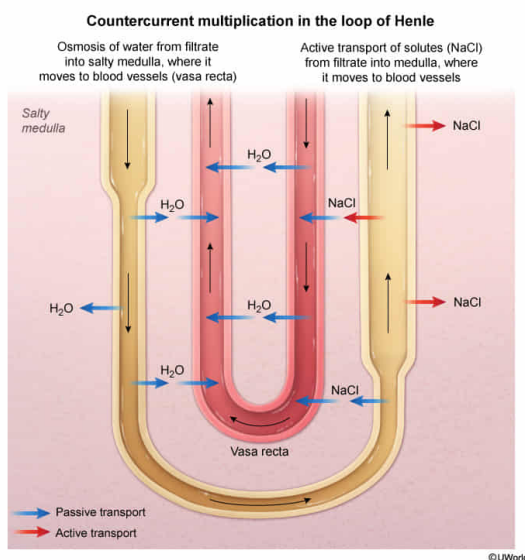
Renal studies in mice have revealed that a mutation in various sodium transporters causes impaired active transport of NaCl from the filtrate within the loop of Henle to the medulla. If mice with wild-type sodium transporters were compared to mice with the mutated sodium transporters, the mutant mice would likely exhibit all of the following EXCEPT:

- ☐ A. decreased saltiness of the medulla. [10%]
- ✓ ☒ B. decreased urine output. [60%]
- ☐ C. decreased renal water reabsorption. [13%]
- ☐ D. decreased blood pressure. [15%]

Correct

61% answered correctly

Explanation:



Each **kidney**, divided into an outer cortex and inner "salty" medulla, contains many functional units known as nephrons. In the renal cortex, blood is first filtered through **glomerular capillaries** into the Bowman's capsule (a cup-shaped structure) of each nephron. The filtered fluid (filtrate) then flows through the **tubular segments** of the nephron. The proximal tubule is the first segment the filtrate traverses. Next, the filtrate flows into the loop of Henle, a *two-limbed* structure *stretching into the medulla* that reabsorbs water and salt (NaCl) from the filtrate into the body via **countercurrent multiplication**:

1. Filtrate flows through the descending limb of the loop, which is *highly* permeable to water but *impermeable* to NaCl. Water will move from areas of low solute concentration to areas of high solute concentration. Accordingly, because salt concentration in the medulla is *high*, water is passively reabsorbed (via osmosis) from the filtrate flowing through the descending limb into the salty medulla, where it is taken up by blood vessels.
2. Filtrate then flows into the ascending limb, which is *highly* permeable to NaCl but *impermeable* to water. Initially, NaCl is passively reabsorbed into the medulla as filtrate moves up the ascending limb. However, as the ascending limb nears the cortex, NaCl is **actively transported** from the filtrate into the medulla, preserving the medulla's high salt concentration. Because **water follows NaCl**, the saltiness of the medulla promotes

continued water reabsorption from the descending loop of Henle and the collecting duct.

Following passage through the loop of Henle, filtrate travels through the distal tubule into the collecting duct, where water and salt reabsorption continue. Remaining filtrate flows to the **bladder** and is excreted from the body as urine.

Studies in mice indicate that sodium transporter mutations **impair active transport of NaCl** from the loop of Henle into the medulla. Accordingly, compared to wild-type mice, mice with mutated sodium transporters would exhibit a **decrease** in both medullary salt concentration and water reabsorption (**Choices A and C**), a **decrease** in blood pressure (**Choice D**), and an **increase** in urine output (**Choice B**).

Educational objective:

In the loop of Henle, active transport of NaCl from filtrate in the ascending limb maintains the saltiness of the renal medulla, facilitating passive water reabsorption from filtrate in the descending limb and collecting duct.

Time spent:0 sec

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